

Rahul Kumar Singh

AI & Machine Learning Engineer

Pune, India | +91-8809109905 | singh205184@gmail.com | [GitHub](#) | [LinkedIn](#) | [Portfolio](#)

Summary

AI/ML Engineer with hands on experience building end-to-end AI systems spanning computer vision, agentic LLM applications, and real-time distributed systems. Proficient in training and deploying deep learning models (YOLOv8/YOLOv11, FaceNet), building multi-agent LLM pipelines with LangGraph and RAG, and engineering full-stack, WebSocket driven backends with FastAPI, Node.js, and MongoDB.

Technical Skills

Languages	Python, C++, JavaScript, SQL
AI / ML	TensorFlow, Scikit-learn, OpenCV, YOLOv8, NumPy, Pandas
Generative AI	LangGraph, RAG, ChromaDB, Pinecone, OpenAI API, Hugging Face
Backend	FastAPI, Flask, Node.js, Express.js, REST APIs, WebSockets
Frontend	React.js, Next.js
Databases	MongoDB, Firebase
Tools	Git, GitHub, Docker, Streamlit, Hugging Face, Vercel, Render

Work Experience

Machine Learning Intern <i>RVNS Solutions Pvt. Ltd.</i>	Feb 2026 – May 2026 <i>Remote</i>
- Designed and deployed an AI-Based Oral Health Analysis System using YOLOv8 capable of detecting 5 oral diseases from mouth images with real-time color coded annotations. - Built full-stack architecture using React.js, Node.js, Express.js, Flask API, MongoDB Atlas with authentication, prediction history dashboard, and downloadable reports. - Integrated custom-trained YOLOv8 model with Flask API and deployed the complete system on Vercel, Render, and Hugging Face Spaces for cloud-based real-time inference.	

Projects

AI-Based Oral Health Analysis System — <i>Full-Stack AI Healthcare, YOLOv8</i>	Live Demo	GitHub	2026
- Built and deployed a full-stack diagnostic app that detects and classifies 5 oral conditions (Cancer, Calculus, Caries, Gingivitis, Ulcer) using a custom-trained YOLOv8 model, rendering each class with a distinct color-coded bounding box. - Architected a three-tier pipeline (React.js → Node.js/Express → Flask) with prediction history, authentication, and downloadable reports, persisting results and confidence scores in MongoDB Atlas. - Deployed independently across Vercel, Render, and Hugging Face Spaces for scalable, cloud-based real-time inference.			
AgentVerse — Multi-Agent AI Chat Platform — <i>LangGraph, FastAPI, Next.js, RAG</i>	GitHub		2026
- Developed a full-stack agentic AI chat framework using LangGraph for behavior orchestration, a fully async FastAPI backend, and Next.js for a streaming (SSE) chat interface. - Implemented RAG-based knowledge retrieval with ChromaDB and a locally-hosted bge-m3 embedding model for custom knowledge base Q&A. - Designed a multi-agent supervisor system coordinating math, code-generation, and general-purpose agents, with custom tool invocation and persistent chat history.			
Drone Surveillance System — <i>YOLOv11, FaceNet, WebSockets, PyBullet RL</i>	GitHub		2025
- Built an enterprise-style real-time surveillance platform combining autonomous drones and CCTV feeds, running YOLOv11 detection, FaceNet recognition, and crowd-density analytics on live video. - Engineered a FastAPI + WebSocket communication server for bidirectional drone/application messaging, alert lifecycle management, and MongoDB Change Streams-based real-time sync across clients. - Built a physics based test environment with PyBullet and Stable-Baselines3 (PPO) for RL-driven autonomous drone navigation and alert validation, without physical hardware.			

Education

Dr. D. Y. Patil Institute of Engineering, Management & Research (DYPIEMR)	2023 – 2027
B.E. in Artificial Intelligence & Data Science, Pune	CGPA: 8.55

Certifications

- Oracle Cloud Infrastructure 2025 Certified Generative AI Professional** — Oracle
- Model Context Protocol: Advanced Topics** — Anthropic Academy
- Career Essentials in Generative AI** — Microsoft & LinkedIn Learning